Design Procedure for a 2.45 GHz 1x4 Microstrip Patch Antenna Array Using CST Studio Suite

Step 1: Create a New Project in CST Studio Suite

1. Launch **CST Studio Suite (Learning Edition)**.
2. Click **New Project**.
3. Under **Application**, select **Microwave & RF / Optical**.
4. Select **Antennas** as the project type.
5. Choose **Planar Antenna** as the antenna template.
6. Select the solver:
    - **Time Domain Solver** (recommended), or
    - **Frequency Domain Solver**.
7. Keep the default unit settings:
    - **Length:** mm
    - **Frequency:** GHz
    - **Time:** ns
8. Set the simulation frequency range:
    - **Minimum Frequency:** 0 GHz
    - **Maximum Frequency:** 5 GHz
9. Enter the **center frequency** as **2.45 GHz**.
10. Click **Finish** to create the project and open the CST modeling environment

Step 2: Design Calculation Reference

The dimensions are all Calculate Theoricatily and a separate calculation document has been prepared containing the detailed mathematical derivations, design equations, substitution of values, and optimized antenna parameters used during the design process.

The reference document includes:

- Patch width calculation
- Free Space Wavelength
- Effective dielectric constant
- Fringing field extension
- Effective and actual patch length calculations
- Ground plane dimension calculations
- Substrate Dimensions
- Guided Wavelength
- Element Spacing
- Feed line length Calculations
- Quarter-wave Transformer Calculations
- Feed width and feed length calculations
- Inset feed optimization
- Corporate Feed Network

  Final optimized design parameters and CST simulation results for **2.45 GHz 1x4 Microstrip Array Antenna**

Design Calculation PDF

Step 3: Design of the Ground Plane

The ground plane is the first component created in the antenna model as it serves as the reference conducting surface for the microstrip patch antenna.
In CST Studio Suite 2026, the ground plane is modeled as a rectangular brick positioned at the bottom of the antenna structure.
Initially, a new **Brick** is created and renamed as **GROUND**.Enter the Dimensions as per the Design variables. The X and Y coordinates are specified based on the calculated substrate dimensions, while the Z coordinates represent the thickness of the copper layer. The material assigned to the ground plane is **Copper (annealed)**, which provides high electrical conductivity and minimizes conductor losses.
The ground plane is positioned at the base of the antenna structure with a thickness of **0.035 mm**
The completed ground plane acts as the foundation for constructing the remaining layers of the microstrip patch antenna array.
Ground Plane Parameters Used:

1. **Object Name: GROUND**
2. **Material Name:Copper**

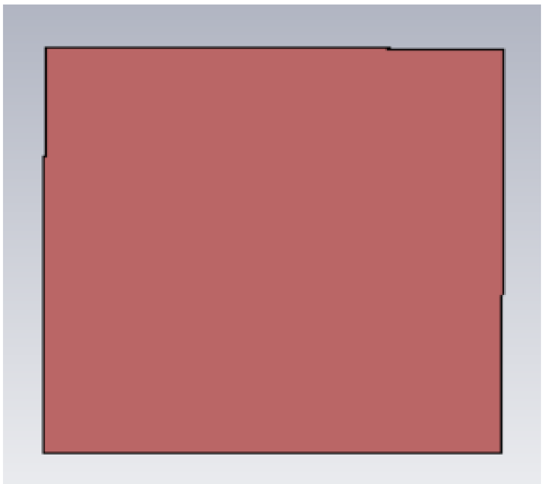

Figure 3.1: Ground Plane Created in CST Studio Suite.

Step 4: Create the Substrate

After creating the ground plane, the dielectric substrate is modeled above it.
The substrate provides mechanical support to the radiating patch and electrically separates the patch from the ground plane.Therefore, selecting an appropriate substrate material and thickness is essential for achieving the desired antenna performance.

In CST Studio Suite 2026, a new **Brick** is created and renamed as **SUBSTRATE**.
The substrate is assigned the material **FR-4 (lossy)**, which has a relative dielectric constant (εr) of approximately 4.3.The X and Y dimensions of the substrate are chosen to match those of the ground plane, ensuring complete support for the antenna array.

After entering the required dimensions and material properties, the geometry is previewed to verify its placement and dimensions before confirming the model.

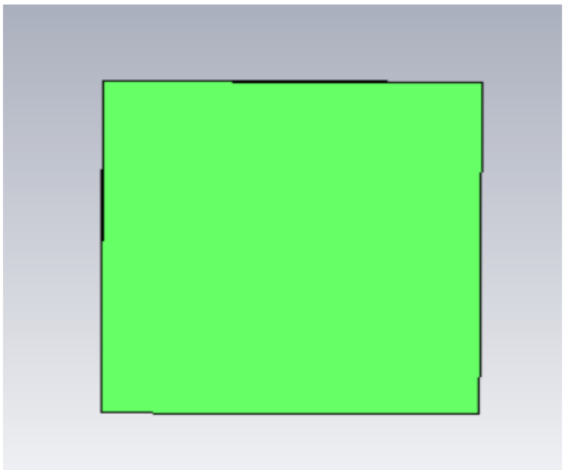

Figure 4.1: Substrate Created in CST Studio Suite

Step 5: Design of the Radiating Patch and Feedline

After creating the substrate, the radiating patch is designed on the top surface of the substrate.

(a) Creation of the Rectangular Patch

Initially, a rectangular Brick is created and renamed as **PATCH**. The patch is placed centrally on the upper surface of the substrate using the calculated dimensions obtained from the design equations. The material assigned to the patch is **Copper (annealed)** because of its high electrical conductivity and low conductor loss. The patch is positioned directly above the substrate by defining the Z-coordinates corresponding to the substrate thickness and copper thickness. After entering the required dimensions,the geometry is previewed and confirmed.

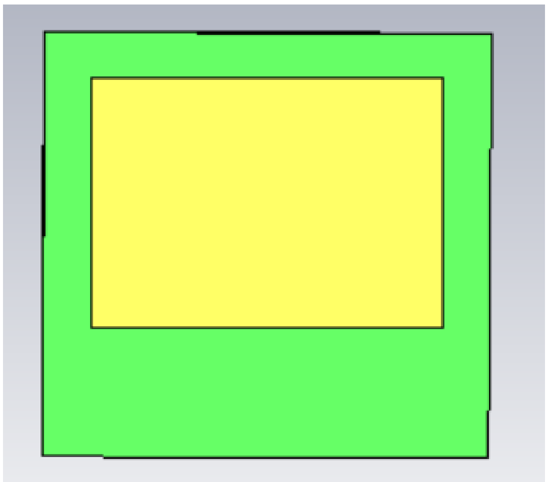

Figure 5.1: Rectangular Patch Created in CST Studio Suite.

(b) Creation of the Inset Feed Cut

To obtain proper impedance matching, an **inset feed slot** is introduced at the center of the lower edge of the patch. A small rectangular brick named **INSET_FEED_CUT** is created with dimensions corresponding to the calculated inset depth and slot width. This temporary brick is assigned a different material only for modeling purposes. Using the **Boolean → Cut Away Highlighted Shape** operation, the inset feed slot is removed from the patch, producing the desired inset-fed geometry. The notch allows the feedline to excite the patch at a location where the input impedance is approximately **50 Ω**, thereby minimizing impedance mismatch.

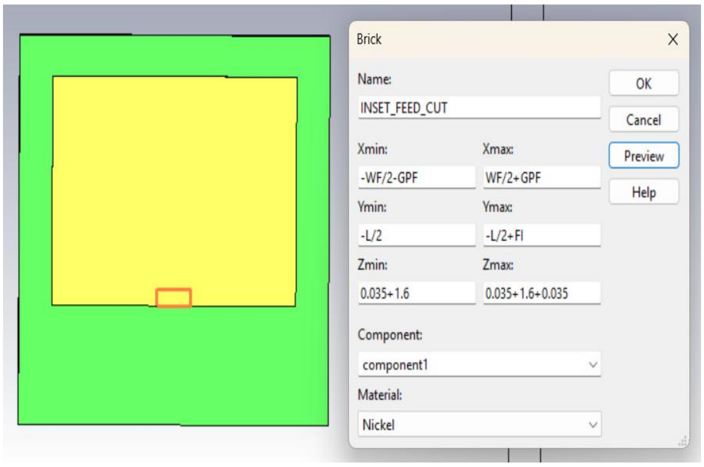

Figure 5.2: Inset Feed Created in CST Studio Suite.

(c) Creation of the Feedline

After creating the inset slot, a microstrip feedline is designed to excite the patch. A rectangular brick named **FEEDLINE** is created and connected to the inset feed slot. The feedline is designed with the calculated width corresponding to a **50 Ω characteristic impedance**, ensuring efficient power transfer from the input port to the radiating patch. The feedline extends from the lower edge of the substrate to the inset notch and is placed on the same conducting layer as the patch.

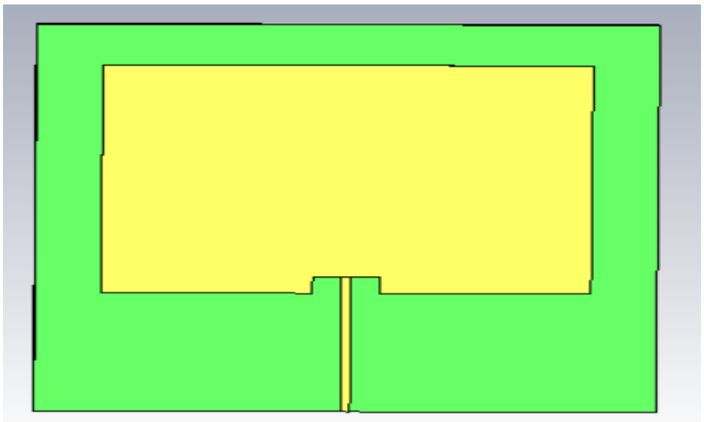

Figure 5.3: Feedline Created in CST Studio Suite.

(d) Merging the Patch and Feedline

After creating the feedline, it is electrically connected to the radiating patch using the **Boolean → Add** operation. This operation merges the patch and feedline into a single conducting structure, ensuring electrical continuity between the feed network and the radiating element. The completed geometry consists of an inset-fed rectangular microstrip patch with a 50 Ω feedline, forming the fundamental radiating element that will later be replicated to construct the **1 × 4 microstrip patch antenna array**.

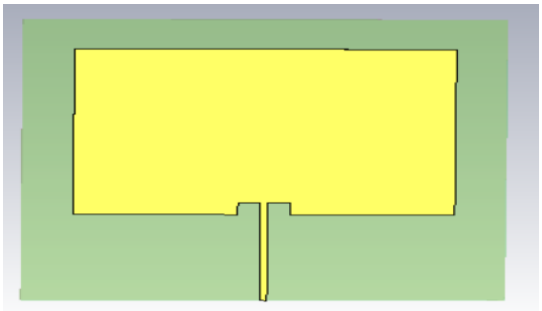

Figure 5.4: Patch Created in CST Studio Suite

Step 6: Creation of the Second Patch Element

After completing the design of the first inset-fed microstrip patch antenna, the next step is to create the second radiating element of the array. Instead of redesigning the patch from the beginning, the already designed patch along with its feedline is duplicated using the **Transform** operation available in CST Studio Suite 2026. This approach ensures that both antenna elements have identical dimensions, feed structures, and electrical characteristics, thereby maintaining symmetry in the antenna array. The spacing between adjacent patches is carefully selected to minimize mutual coupling while achieving the desired array gain and radiation performance.

The completed first patch is selected and translated along the X-axis by the calculated inter-element spacing. During this process, the **Copy** option is enabled so that the original patch remains unchanged while an identical second patch is created. After applying the translation, the geometry is verified to ensure that both patches are correctly aligned and separated by the required distance.

Procedure

1. Select the completed Patch together with its inset feed and feedline.
2. **Open Transform → Translate →Copy → Set the Translation Vector only along the X-axis → Preview → OK**
3. Verify that the newly created patch has identical dimensions, inset feed, and feedline as the original patch.
4. Ensure that sufficient spacing exists between the two radiating elements to avoid physical overlap and to maintain the required inter-element spacing for proper array operation.

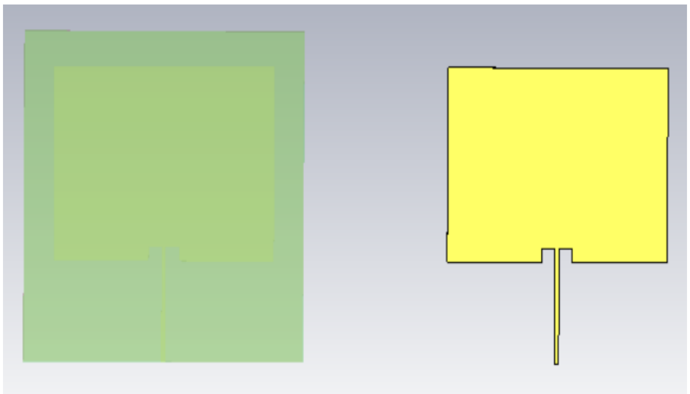

Figure 6.1: Second Patch Created in CST Studio Suite

Step 7: Design of the Interconnecting Feedline

After creating the two identical patch elements, the next step is to design the **interconnecting feedline** that electrically connects both antenna elements. The feedline is drawn horizontally to connect the vertical feedlines of the two patch elements. The feedline is placed on the same conducting layer as the patches.

The dimensions of the feedline are defined using the design variables so that the spacing between the two patches and the feed network can be easily modified during optimization. After entering the required dimensions, the geometry is previewed to verify that the feedline correctly joins the two antenna elements without any discontinuity.

Procedure:

1. Create a Brick and rename it as **FEEDLINE2**.
2. Assign the material as **Copper (annealed)**.
3. Enter the dimensions as per the design.
4. Click **Preview** to verify the geometry.
5. Confirm that the horizontal feedline is properly aligned with both vertical feedlines.
6. Click **OK** to create the feedline.

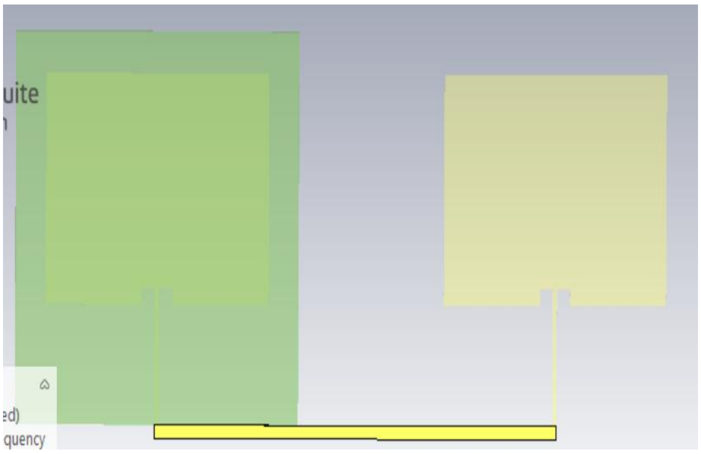

Figure 7.1: Interconnecting Created in CST Studio Suite

Step 8: Design of the Main Input Feedline

After designing the horizontal interconnecting feedline, the next step is to create the main input feedline that supplies RF power from the excitation port to the corporate feed network. This vertical feedline serves as the primary transmission path between the input port and the first-stage power divider. It is designed with the appropriate width to maintain the required characteristic impedance, ensuring efficient power transfer with minimum reflection. The feedline is positioned at the center of the horizontal feedline so that the input power is equally distributed to both patch elements.
In CST Studio Suite 2026,
Procedure

1. Create a new **Brick** and rename it as **FEEDLINE3**.
2. Assign the material as **Copper (annealed)**.
3. Position the feedline vertically below the center of **FEEDLINE2**.
4. Ensure that the feedline is aligned with the midpoint of the horizontal feedline to achieve equal power division.
5. The dimensions are defined using the calculated design variables
6. Click **Preview** to verify the location and dimensions.
7. Confirm that the feedline is properly connected to the center of the horizontal feedline without any gaps.
8. Click **OK** to complete the creation of the feedline.
9. Inspect the model to ensure proper alignment and electrical continuity with the existing feed network.

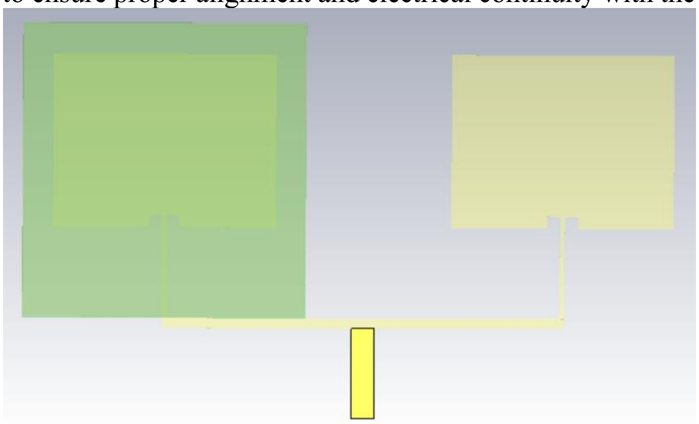

Figure 8.1: Second Patch Created in CST Studio Suite

Step 9: Modification of the Ground Plane and Substrate Dimensions

After completing the design of the two-element corporate feed network, the dimensions of the ground plane and substrate are increased to accommodate the second patch element and the complete feeding structure

In CST Studio Suite 2026, the existing **GROUND** and **SUBSTRATE** bricks are edited instead of creating new

objects. The **Xmax** and **Ymin** coordinates are modified according to the total width of the two-element array and the extended feed network. The material properties remain unchanged, with **Copper (annealed)** assigned to the ground plane and **FR-4 (lossy)** assigned to the substrate. After updating the dimensions, the geometry is previewed to ensure that the enlarged ground plane and substrate completely cover both patch elements and the corporate feed network.
Procedure:

1. **GROUND**

Brick → GROUND →Copper(annealed) → Enter the Dimensions → Preview →Ok

2. **SUBSTRATE**

Brick → GROUND →Copper(annealed) → Enter the Dimensions → Preview →Ok

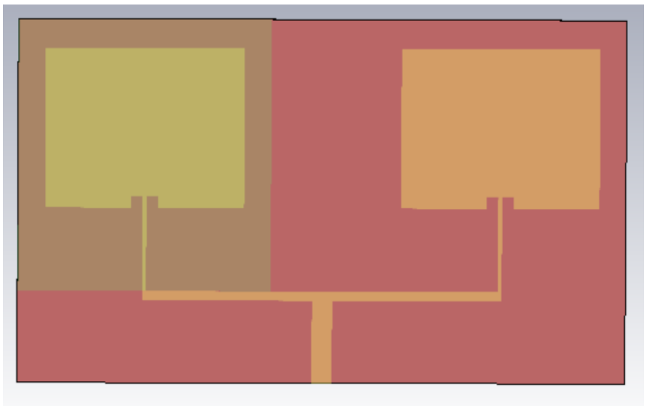
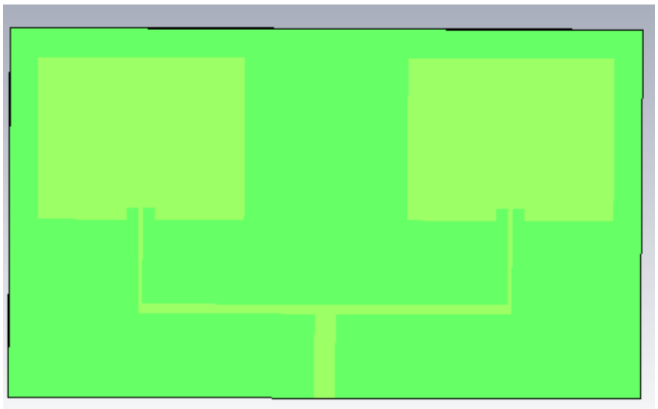

Figure 9.1: Modified Ground Created in CST Studio Suite **Figure 9.2:** Modified Substrate Created in CST Studio Suite

Step 10: Merging the Patch Elements and Feed Network

After creating the two radiating patch elements and completing the corporate feed network, the next step is to merge all the conducting parts into a single conductive structure. This is accomplished using the **Boolean → Add** operation in CST Studio Suite 2026.

In this stage, the first patch is selected as the reference object, and the remaining conducting elements, including the second patch and all feedlines, are added to it. First select the main **PATCH → Modeling → Boolean → Add → PATCH_1 → FEEDLINE2 → FEEDLINE3 →Enter(or click OK).**
After the merging process, the geometry is inspected to confirm that all conducting elements are perfectly connected and no gaps or overlaps exist.
Save the updated antenna model before proceeding to the next design step.

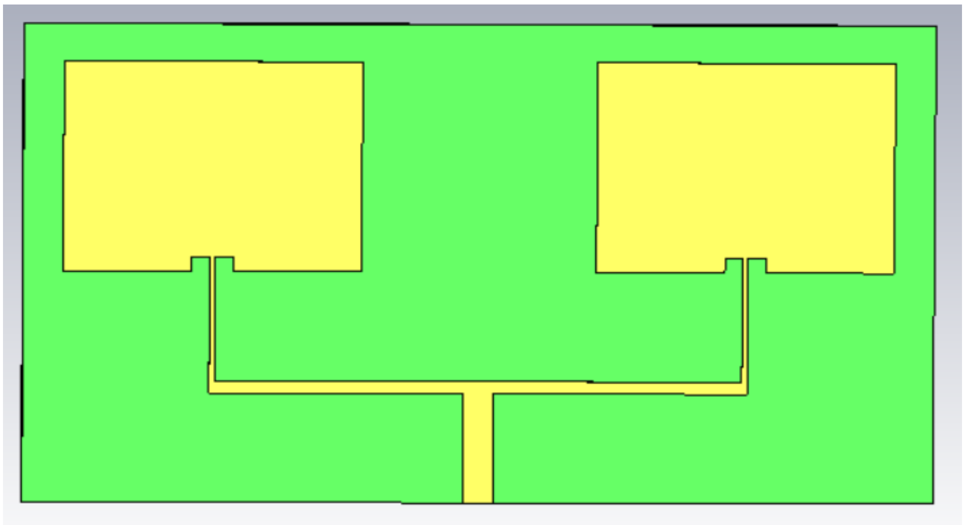

Figure 10.1: 1x2 Microstrip Array Created in CST Studio Suite

Step 11: Replication of the 1 × 2 Antenna Array to Form the 1 × 4 Array

After completing the **1 × 2 microstrip patch antenna array**, the next step is to duplicate the entire structure to obtain a **1 × 4 microstrip patch antenna array**. Instead of creating two additional patch elements individually, the complete 1 × 2 array, including the radiating patches and the corporate feed network, is copied using the **Transform → Translate** operation.

In CST Studio Suite 2026, the entire **1 × 2 antenna array** is selected and translated along the **X-axis** while enabling the **Copy** option. A new parameter **D1** is defined to represent the spacing between the two 1 × 2 sub-arrays. The translation distance is specified using the design variables so that the spacing between the two sub-arrays can be easily modified during optimization. After previewing the translated geometry, the duplicated structure is confirmed, resulting in four identical patch elements arranged in a linear configuration.

Procedure

1. Select the complete **1 × 2 antenna array**, including the radiating patches and the feed network.
2. Open **Transform → Translate**.
3. Enable the **Copy** option.
4. Set the **Repetition Factor** to **1**.
5. Define a new parameter:
    a. **Parameter Name:** D1
    b. **Value:** 14.58 mm(or any other specified value according to the design)
6. Specify the translation vector
7. Click **Preview** to verify the location of the duplicated array.
8. Ensure that the spacing between the two 1 × 2 arrays is uniform and that no overlap occurs.
9. Click **OK** to complete the translation.
10. Verify that four identical patch elements are created with the same feed network configuration.

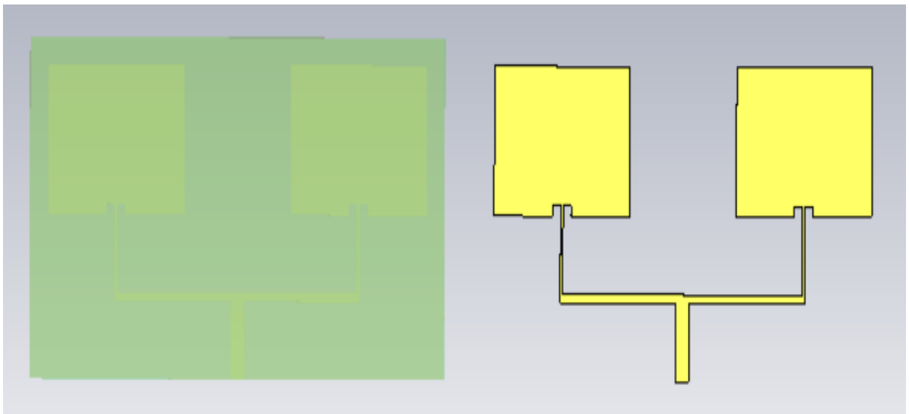

Figure 11.1: Replication of 1x2 Patch Created in CST Studio Suite

Step 12: Design of the Interconnecting Feedline Between the Two 1 × 2 Sub-Arrays

After replicating the **1 × 2 antenna array** to obtain the four radiating patch elements, the next step is to interconnect the two sub-arrays using a horizontal microstrip feedline. This feedline forms the second stage of the corporate feeding network and provides a common transmission path between the left and right **1 × 2** antenna sections

In CST Studio Suite 2026, a new **Brick** is created → renamed as **FEEDLINE5**. The feedline is drawn horizontally between the two **1 × 2** antenna arrays and aligned with the lower feed network. The material assigned to the feedline is **Copper (annealed)**. The dimensions are defined using the design parameters so that the spacing between the two sub-arrays can be easily adjusted. After entering the required dimensions, the geometry is previewed and verified before confirming the design.

Procedure:

Create a new **Brick** → rename it as **FEEDLINE5** → Assign the material as **Copper (annealed)**→ Enter the dimensions using the calculated design variables → Click **Preview** to verify the geometry → Click **OK**

to complete the creation of the feedline.
Inspect the model to ensure there are no gaps or overlaps between the newly created feedline and the existing corporate feed network.

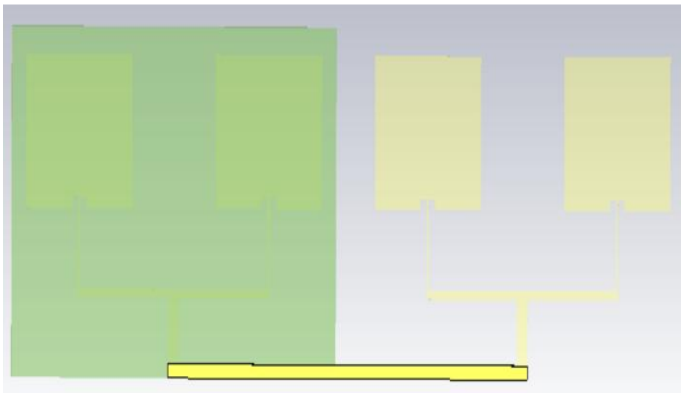

Figure 12.1: Interconnecting Feedline Created in CST Studio Suite

Step 13: Design of the Main Input Feedline for the 1 × 4 Corporate Feed Network

After connecting the two **1 × 2 antenna sub-arrays** using the horizontal feedline, the next step is to design the final input feedline for the complete **1 × 4 microstrip patch antenna array**. The feedline is designed to maintain the required characteristic impedance and ensure equal power distribution to all four radiating patch elements.

In CST Studio Suite 2026, a new **Brick** is created to represent the final vertical input feedline. The feedline is positioned at the Center of the horizontal interconnecting feedline and assigned the material **Copper (annealed)**. The dimensions are specified using the calculated design variables. After entering the required parameters, the geometry is previewed and verified before confirming the design.

Procedure

Create a new **Brick** → rename it as **FEEDLINE** → Assign the material as **Copper (annealed)** → Enter the dimensions using the calculated design variables → Click **Preview** to verify the geometry → Click **OK** to complete the creation of the feedline.

Inspect the geometry to ensure complete electrical continuity throughout the corporate feed network.

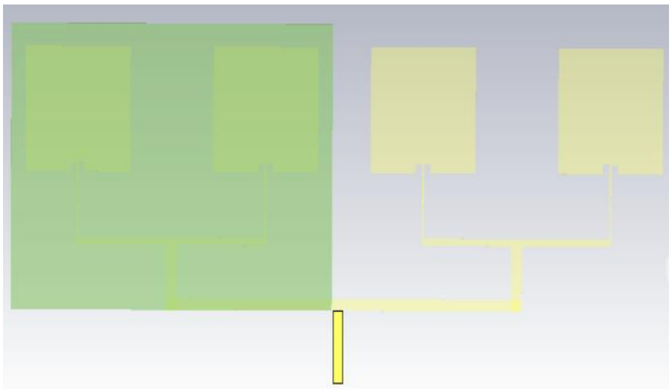

Figure 13.1: Main Input Feedline Created in CST Studio Suite

Step 14 : Modification of the Ground Plane and Substrate for the 1 × 4 Antenna Array

After completing the corporate feeding network for the **1 × 4 microstrip patch antenna array**, the dimensions of the ground plane and substrate are increased once again to accommodate all four radiating patch elements and the complete feed network.

In CST Studio Suite 2026, the existing **GROUND** and **SUBSTRATE** objects are modified instead of creating new geometries.The **Xmax** and **Ymin** coordinates are updated according to the total dimensions of the 1 × 4 antenna array and the extended corporate feed network. The material properties remain unchanged, with **Copper (annealed)**

assigned to the ground plane and **FR-4 (lossy)** assigned to the substrate. After modifying the dimensions, the geometry is previewed to verify that all four patch elements and the complete feed network are fully enclosed before confirming the design.
Procedure:

1. **GROUND**
   Brick → GROUND →Copper(annealed) → Enter the Dimensions → Preview →Ok
2. **SUBSTRATE**
   Brick → GROUND →Copper(annealed) → Enter the Dimensions → Preview →Ok

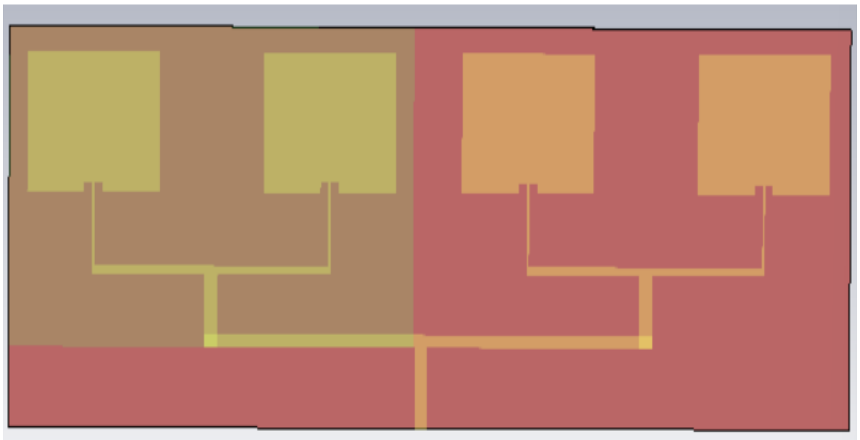

Figure 14.1: Modified Ground Created in CST Studio Suite

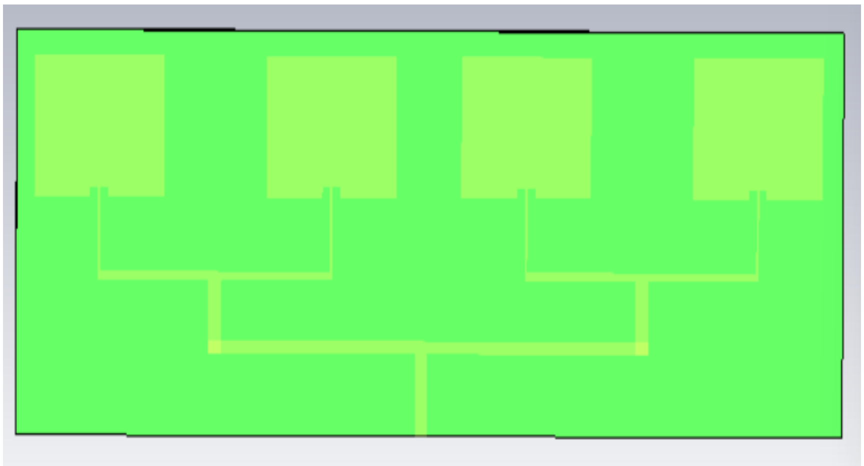

Figure 14.2: Modified Substrate Created in CST Studio Suite

Step 15: Merging All Conducting Elements to Form Complete 1 × 4 Array

After completing the geometry of the **1 × 4 microstrip patch antenna array**, the final step in the design process is to merge all the radiating patches and feedlines into a single conducting structure. Since the antenna has been created using multiple copper bricks, these individual conductors must be combined to establish continuous electrical connectivity throughout the corporate feed network. This is achieved using the **Boolean → Add** operation in CST
Procedure

1. Select the main **PATCH** object from the Components tree.
2. Open **Modeling → Boolean → Add**.
3. Select all remaining copper conducting objects like patches and feedlines one by one
4. Press **Enter** or click **OK** to perform the Boolean Add operation.
5. Verify that all selected conductors are merged into a single object.
6. Inspect the geometry carefully to confirm that there are no discontinuities, overlaps, or unwanted gaps.
7. Save the completed antenna model before proceeding to port definition and simulatioion.

Step 16: Waveguide Port Definition and Port Extension Calculation

After completing the geometry of the **1 × 4 microstrip patch antenna array**, the excitation port is defined at the input feedline., the **Port Extension Coefficient** is calculated using the **CST Port Extension Macro**. This automatically determines the optimum port dimensions based on the feedline width, substrate thickness, dielectric constant, and operating frequency range.
In CST Studio Suite 2026,

1. Select the **metal face** at the end of the main input feedline.
2. Press **F** to zoom and accurately select the feedline face.
3. Open **Home → VBA Macros → Solver → Ports → Calculate Port Extension Coefficient**.
4. Select the transmission line type as **Microstrip**.
5. Enter the required parameters:
   - **Feedline Width (W)**
   - **Substrate Thickness (h)**
   - **Relative Dielectric Constant (εr)**
   - **Frequency Range:**
6. Click **Calculate** to determine the optimum port extension coefficient.
7. Verify the calculated **Extension Coefficient (k)**.
8. Click **Construct Port from Picked Face** to automatically create the waveguide port.
9. Close the macro window after the port has been successfully generated.
10. Verify that the waveguide port is correctly attached to the input feedline and fully covers the feed conductor and substrate.

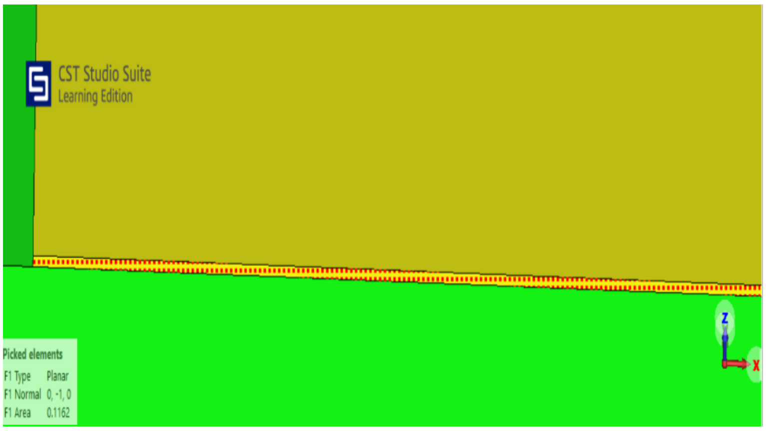

Figure 16.1: Selecting the feed line face

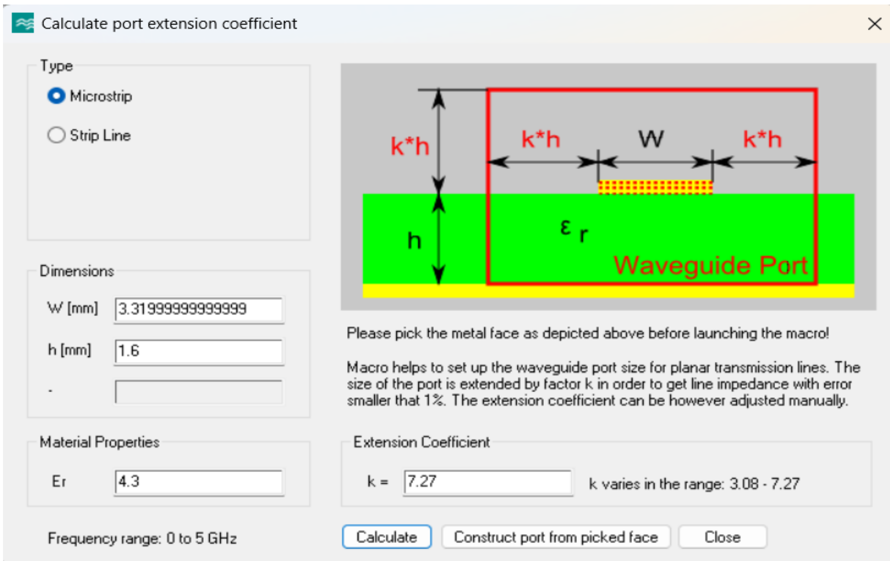

Figure 14.2: Calculate Port Extension Coefficient Dialog

Step 17: Configure the Farfield Monitor and Run the Simulation

The **Farfield Monitor** is used to calculate the antenna radiation characteristics, such as gain, directivity, and radiation pattern, at the desired operating frequency.

Procedure

1. Navigate to **Simulation → Field Monitors**.
2. In the **Monitor** dialog box, select **Farfield/RCS** as the monitor type.
3. Under **Specification**, select **Frequency** and enter the **center frequency** of the antenna (e.g., **2.45 GHz**).
4. Click **Apply**, then click **OK** to create the farfield monitor.
5. After configuring the monitor, navigate to **Home → Start Simulation** to begin the simulation.

Once the simulation is complete, CST Studio Suite automatically generates the electromagnetic results, including **S-Parameters (S11)**, **VSWR**, **Gain**, **Directivity**, **Farfield Radiation Pattern**, and **Surface Current Distribution**, which can be analyzed to evaluate the performance.

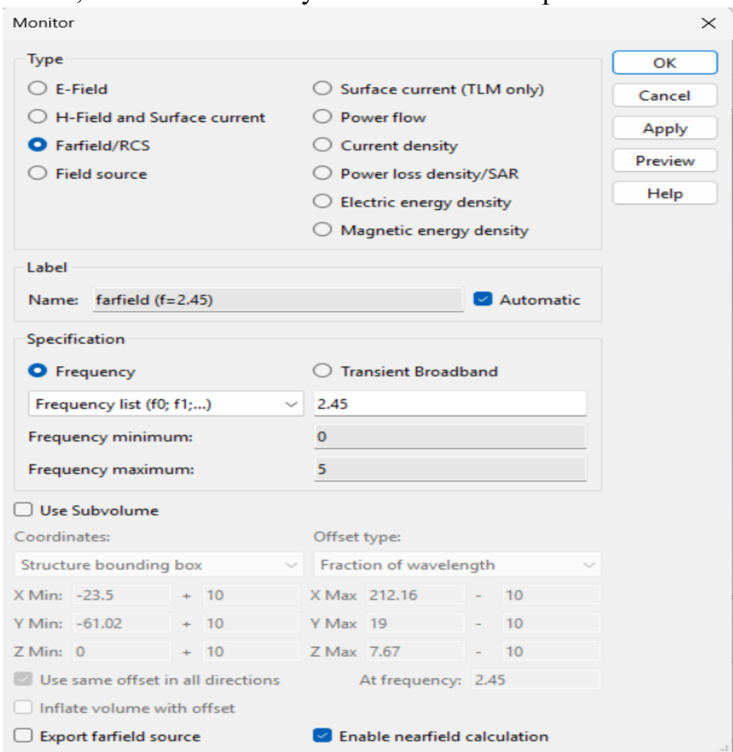

Figure 17.1: Far Field Monitor Configuration

Step 18: View and Analyze the Simulation Results

After the simulation is completed successfully, the results can be viewed from the **Navigation Tree**.

Procedure

1. Expand the **1D Results** folder to view the antenna's input characteristics.
2. Open the following plots:
    - **S-Parameters (S11)** – to analyze the reflection coefficient and resonant frequency.
    - **VSWR** – to evaluate the impedance matching of the antenna.
3. Expand the **Farfields** folder to analyze the radiation characteristics.
4. Open the following results:
    - **3D Radiation Pattern** – to visualize the antenna radiation pattern.
    - **Gain** – to observe the maximum realized gain.
    - **Directivity** – to evaluate the directional characteristics of the antenna.
    - **Polar Plot** – to examine the radiation pattern in different planes.

These results are used to evaluate the overall performance of the antenna array and verify that it meets the desired design specifications.

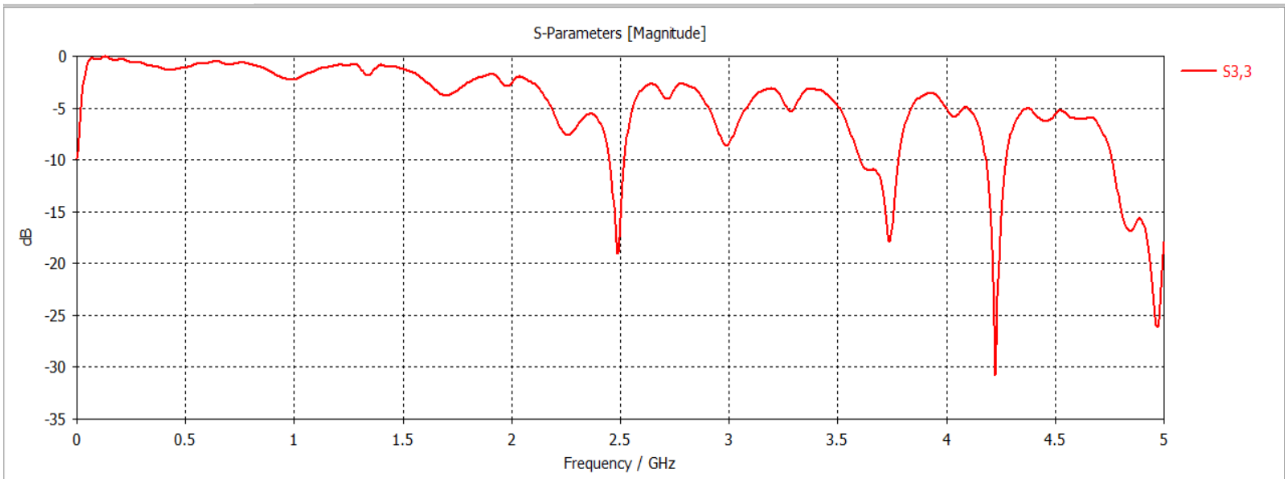

Figure 18.1: S-Parameter(S11)Plot

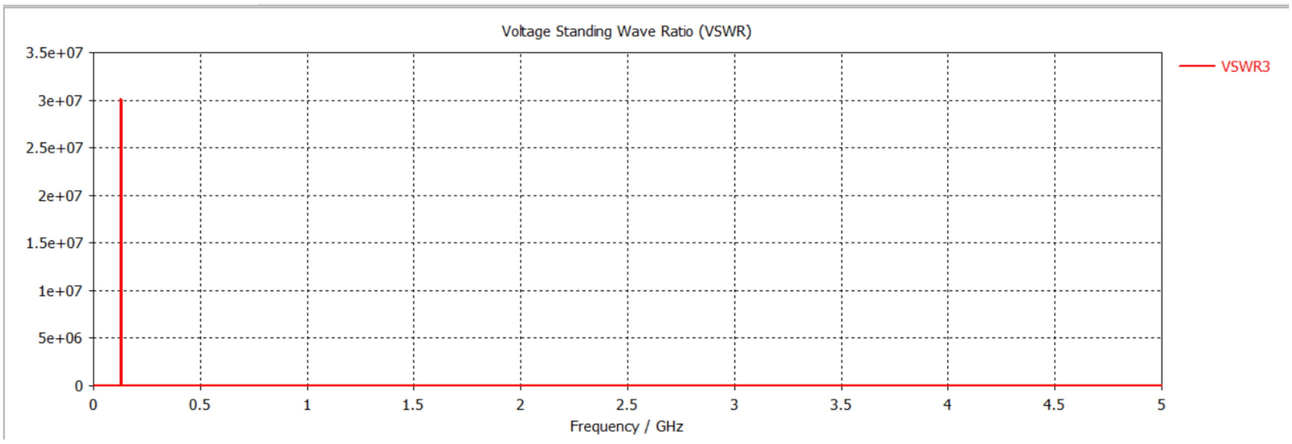

Figure 18.2: VSWR-Plot

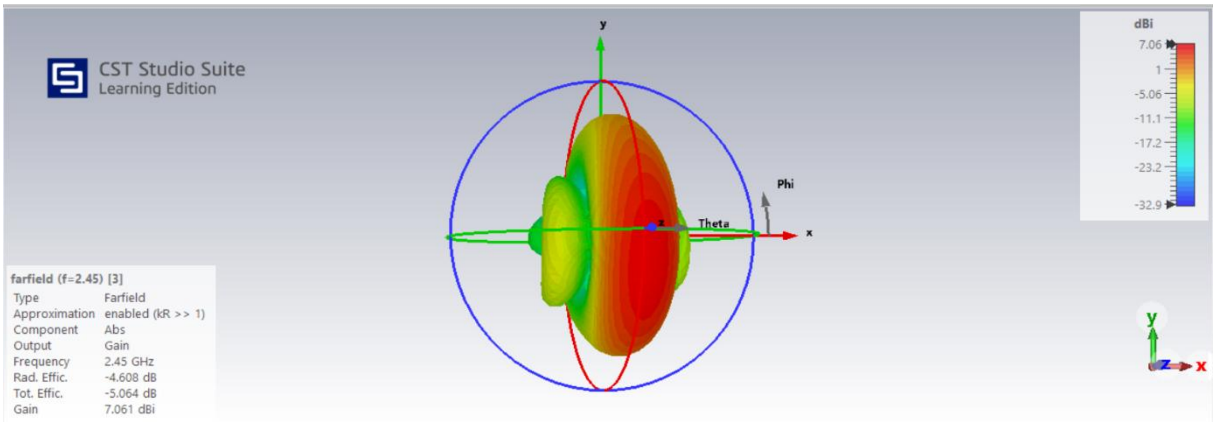

Figure 18.3: Gain Plot

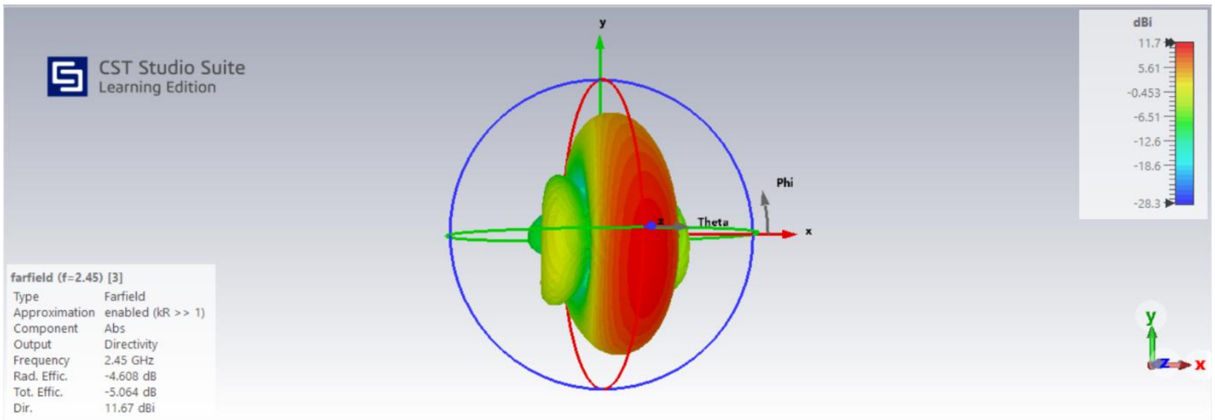

Figure 18.4: Directivity Plot

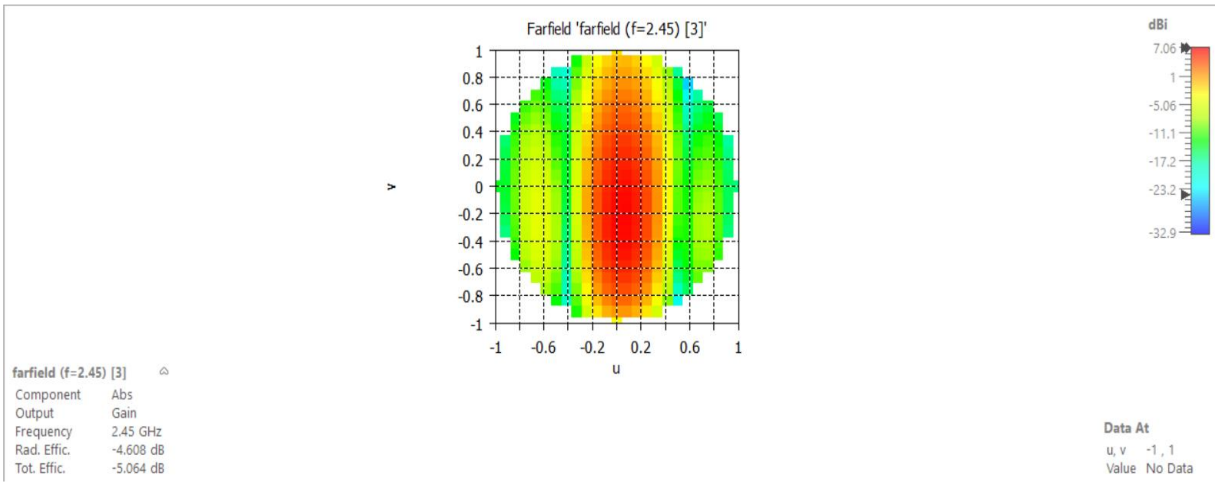

Figure 18.5: 2D-FarField Gain Plot

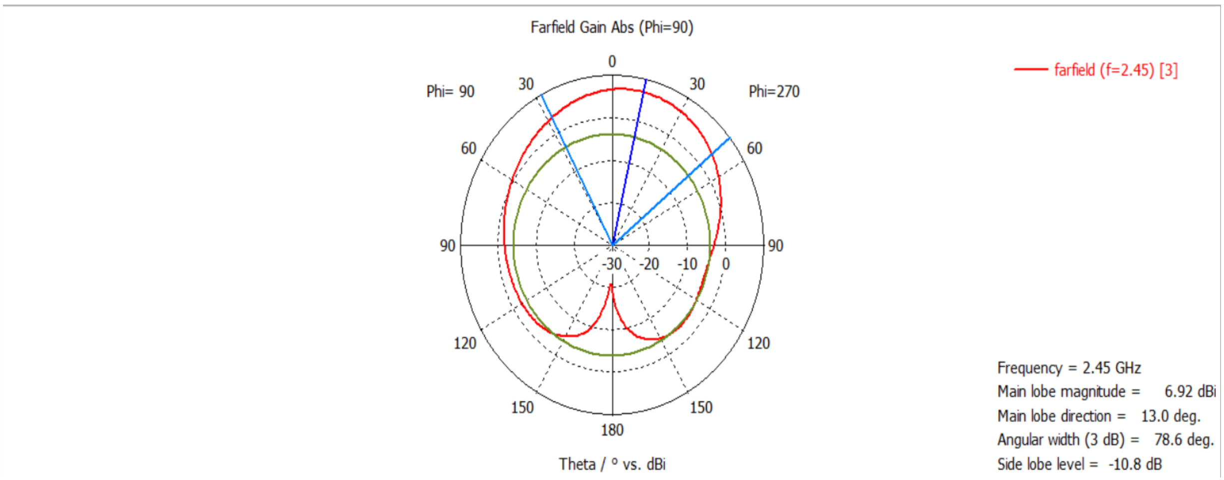

Figure 18.6: Polar Radiation Plot

Final Antenna Model Verification and Parameter Validation

After completing the geometry construction and waveguide port definition, the final **1 × 4 microstrip patch antenna array** is verified and simulated successfully . All four radiating patch elements, the corporate feed network, ground plane, substrate, and waveguide port are carefully inspected to ensure that the antenna geometry is complete and free from modeling errors.

In addition, the design parameters are reviewed to confirm that all dimensions correspond to the calculated values used during the antenna design process. This verification step is essential for obtaining accurate simulation results and avoiding errors during the solver execution.

The complete antenna structure is visually inspected in CST Studio Suite 2026 to ensure that the feedlines are electrically connected to each patch and that no unwanted gaps or overlaps are present. The **Parameter List** is then reviewed to verify the values of all design variables, including the patch dimensions, feedline dimensions, inset feed dimensions, and spacing between adjacent elements. Once the geometry and parameter values are confirmed, the antenna model is considered ready for mesh generation and full-wave electromagnetic simulation.

Parameter List

Name	Expression	Value	Description
WF	= 3.08	3.08	FEED WIDTH
GPF	= 1	1	GAP
FI	= 2	2	INSET FEED LENGTH
L	= 28	28	LENGTH
LF	= 14.58	14.58	LENGTH OF FEEDLINE
W	= 37	37	WIDTH
WF3	= 0.7	0.7	
WF2	= 1.64	1.64	
D1	= 14.58	14.58	

Figure 18.7: CST Parameter List of The Designed Antenna Array

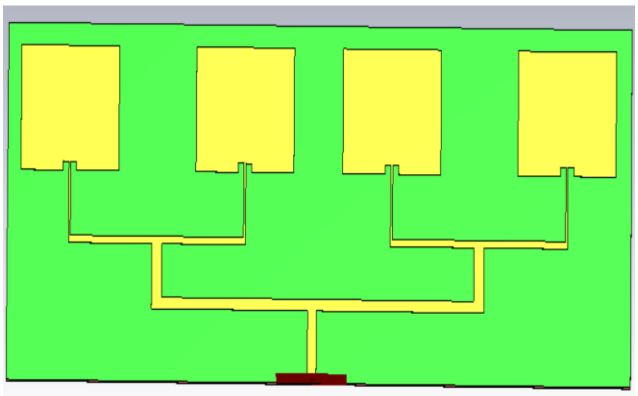
Figure 18.8: Complete Geometry of the 2.45 GHz Microstrip Patch Antenna Array